

Can Karakoc

Email: cankarakoc@berkeley.edu | LinkedIn: [linkedin/can-karakoc](https://www.linkedin.com/in/can-karakoc) | GitHub: [github/can-karakoc](https://github.com/can-karakoc) | Portfolio: can-karakoc.github.io | Work Authorization: Green Card

Software engineer with UC Berkeley Computer Science and Data Science training and experience shipping AI-powered workflows, data pipelines, and production web systems. Strong in Python, JavaScript/TypeScript, SQL, backend services, full-stack development, and ML-enabled products; known for product-minded engineering, fast ownership, and turning ambiguous requirements into reliable user-facing software.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, R, Spark

Backend & Data: APIs, ETL pipelines, feature engineering, PostgreSQL, NoSQL, caching, data validation, workflow automation

Frontend & Product: React, Next.js, Node.js, Streamlit, WordPress, CMS workflows, responsive UI, accessibility, analytics-informed UX

AI/ML Systems: scikit-learn, PyTorch, TensorFlow, embeddings, vector search, FAISS, model evaluation, statistical modeling

Developer Tools: Git/GitHub, Linux, VSCode, Jupyter, Tableau, Power BI, Matplotlib, Plotly

Experience

AI Engineering Intern – Cendra | April 2026 - *Present* | Remote

- Shipped an autonomous AI voice agent for outbound sales workflows, handling lead qualification, CRM correction, callback scheduling, and demo booking in a production startup environment.
- Designed agent workflow logic across enriched lead data, CRM state, post-call classification, WhatsApp/email follow-ups, and human escalation paths.
- Owned product and engineering tradeoffs during a 2-week sprint, prioritizing CRM-native UX, compliance, reliability, and maintainable workflow design over low-leverage features.

Website Content & Design Manager – UC Berkeley Development Engineering | June 2024 - Aug 2025 | Berkeley, CA, USA

- Managed and optimized 20+ production web pages, improving accessibility, information architecture, stakeholder workflows, and content discoverability.
- Built scalable CMS publishing processes that reduced manual update overhead and improved maintainability across multiple program stakeholders.
- Translated product, design, and stakeholder requirements into maintainable web implementations using analytics and user behavior signals to guide iteration.

Data Analysis Intern – Velocity Inc. | May 2023 - Aug 2023 | Istanbul, Turkey

- Built Python data pipelines processing millions of mobile location records for behavioral segmentation and recurring product analytics workflows.
- Engineered 20+ temporal and mobility features from raw data sources, improving downstream model inputs and cohort analysis quality.
- Automated weekly reporting and built Tableau dashboards used by product and strategy teams for operational decision-making.

Projects

Embedding-Based Similarity Search System | Personal Project | App | GitHub

- Built a vector retrieval system that embeds domain-specific sequences and retrieves nearest-neighbor matches using FAISS similarity search.
- Designed a modular backend pipeline for input validation, embedding generation, persistent caching, vector indexing, similarity scoring, and ranked retrieval.
- Implemented cache-hit/miss diagnostics and test workflows to reduce redundant computation, validate retrieval behavior, and support iterative ML-system development.

Energy Production and Carbon Emissions | DATA 102, UC Berkeley | [GitHub](#) | [Paper](#)

- Built predictive models for state-level renewable energy production using GLM, Random Forest, and KNN.
- Developed a causal inference pipeline estimating the impact of electricity prices on carbon emissions.
- Automated model evaluation with cross-validation and bootstrap confidence intervals.

Database Systems | UC Berkeley CS186 | Java

- Implemented multigranularity locking components with hierarchical lock acquisition, promotion, release, and escalation semantics for a database system project.
- Built ARIES recovery logic for forward processing, commit, abort, and end flows, reinforcing transaction safety and systems-level debugging skills.

Education

University of California, Berkeley | Aug 2021 – May 2025

B.A. Computer Science; B.A. Data Science

GPA: 3.8/4.0

Relevant Courses: Designing, Visualizing, and Understanding Deep Neural Networks; Introduction to Machine Learning; Data, Inference & Decisions; Data Engineering; Optimization Models in Engineering; Introduction to Database Systems; Machine Structures; Data Structures.

Leadership

Teaching Assistant – Human-centered Design DeCal | Aug 2023 - May 2023 – Led weekly sessions for 60+ students on design methodology, research synthesis, and prototyping.

Design Consultant – Berkeley Innovation | Jan 2023 - May 2024 – Delivered user-centered design solutions for clients through structured research and iteration.